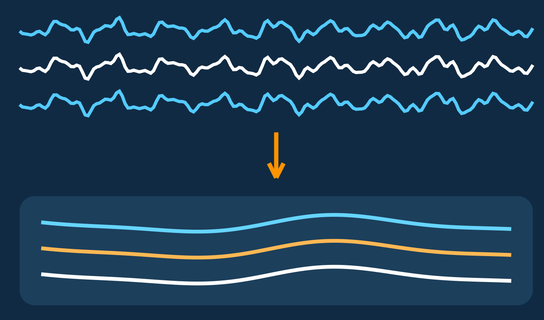
EEG TO BOLD / COMPUTER SCIENCE PHD EXPLORATION

Can EEG recover clinically useful fMRI?

EEG predicts some healthy BOLD. Weak transfer and regional specificity shift the PhD focus to measurement, calibration and reliable abstention.



EXPLORATORY RESEARCH | MONASH M3 EXPERIMENTS | 21 OCTOBER 2026

THE CLINICAL QUESTION

One task. A measurable benefit.

EEG: scalp electrical activity.
BOLD: blood oxygenation in fMRI.
One useful clinical decision?

METHOD / THE ACTUAL M3 PIPELINE

How EEG becomes a prediction

Past EEG is summarized before fitting either model.

24 seconds of EEG
5 scalp groups x 5 frequency bands

↓

300 power features
12 lags; training-only standardization

↓

Ridge or small MLP
Same EEG features; inner tuning

↓

Predict BOLD targets
Score 64 regional series here

MLP learns four target groups. Published models were not reproduced.

01 / EVIDENCE FROM M3

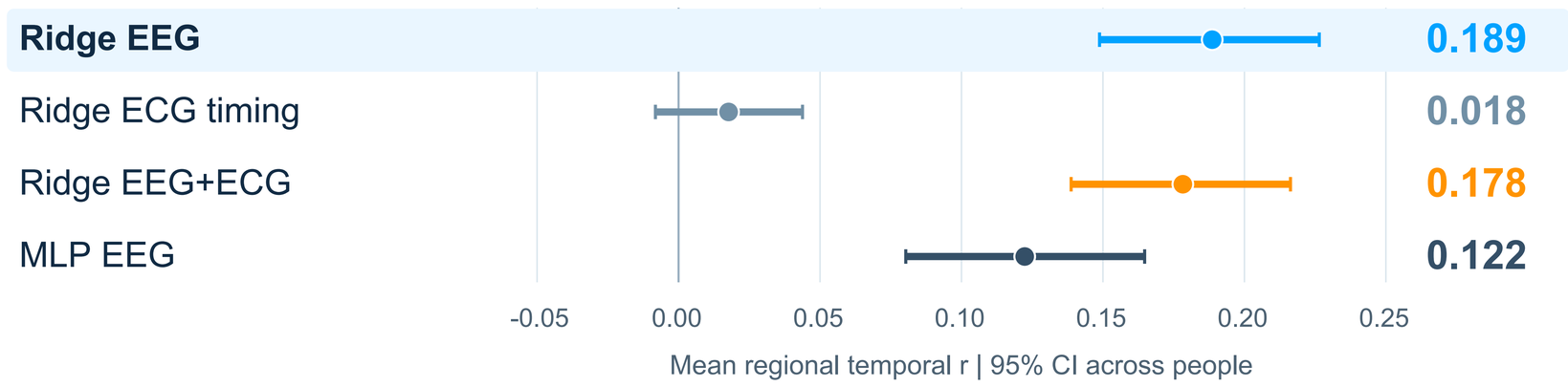
Ridge leads the matched-feature models

ds006040 development | EC-trained / EC-tested | 64 regional targets

27 held-out people

96 fitted experiment cells

300 matched EEG-power features



12 training people x 240 retained target-seconds/person. EEG ridge predictive R^2 = .028.

What does a better prediction mean?

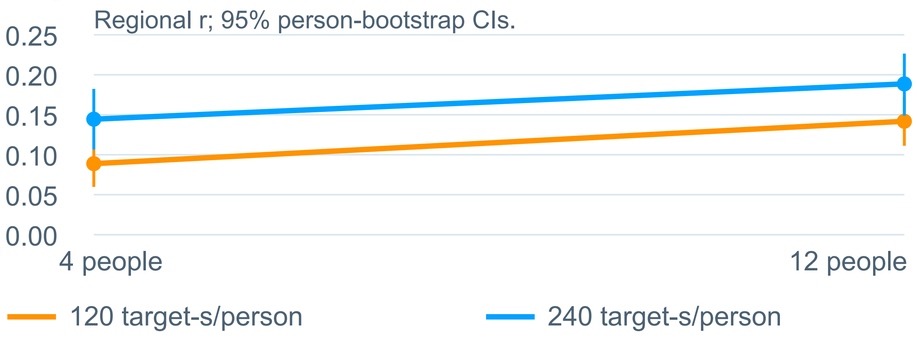
r measures aligned rises/falls; predictive R^2 also penalizes scale and offset errors. EEG ridge r .189, R^2 .028. Regional gain beyond a predicted-global template is uncertain (delta-r .004; 95% CI -.008 to .017).

THE PHD OPPORTUNITY / reliable prediction across people and sites

02 / EFFECT OF TRAINING DATA

People and recording time help

Ridge EEG | same held-out EC evaluation | n=27



METHOD / TEST ON UNSEEN PEOPLE

27 people / 54 EC/EO runs. Three outer folds test unseen people; separate inner validation chooses settings. Compare 4/12 training people and 120/240 retained target-seconds; EEG power, sparse ECG timing or both. One subset order/MLP seed; four equal-weight MLP target groups. The 96 fitted cells reuse these people.

EXTERNAL DATA / SECONDARY EXPLORATION

NATVIEW: transfer remains weak

Strict timing QC: 1/8. Amended QC: 6/8; n=6.
Metadata-only amendment; ingestion pilot excluded.

0.104 regional r

-0.071 predictive R^2

r: 95% CI [-0.0002, 0.207]; includes zero.

740/2304 frames scored (32.1% of all eight).
Frozen VU model/QC; 23 target sites include interpolation.

03 / THREE FALSIFIABLE THESIS QUESTIONS

Recover. Transfer. Know when to stop.

1. **Measure useful information**
   Predict which montage/history/sensors add information; test rankings against cost-matched held-out choices.
2. **Transfer with counted calibration**
   Learn dynamics with missing EEG; test transfer beyond ridge, state templates and output recalibration.
3. **Calibrate when to abstain**
   Compare error at matched coverage. State assumptions: hidden vascular shifts can invalidate reliability.

CUTTING-EDGE MODELS / PROPOSED USES

UnEBOLT [1] | 64 ROIs, r .418 | regional feedback target

UniEFS [2] | 512 ROIs, r .367 | Parkinson classification probe

MARBLE [3] | 64 ROIs, r .434; 2.6M parameters | compact estimation

Healthy temporal r, different protocols: not a ranking or patient validation.

APPLICATIONS / VALIDATION GATES

Wakefulness advisory: compare direct EEG using independent events, false alerts/hour, delay and coverage.
Regional feedback: validate one patient ROI against paired fMRI, then test controlled benefit.
Clinical precedent: Prism's cleared adjunctive PTSD biofeedback excludes diagnostic interpretation [4].
Current limits: retrospective preprocessing, normalized targets, sparse ECG and one development site/seed. Live use and clinical benefit untested [5].

[1] UnEBOLT, MIDL 2026. **[2]** UniEFS, ICML 2026.
[3] MARBLE, MICCAI 2026. **[4]** FDA K222101, 2023.
[5] DECIDE-AI, 2022. Full sources: accompanying report.

Full explanation, methods and code

Seamus Taylor
PhD thesis exploration / Evidence and full methods in the accompanying report